CYBERSECURITY PROJECT DOCUMENTATION

Security Log Analysis & Threat Investigation

Using Splunk SIEM Platform

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Tool: Splunk Enterprise 10.0.1

1. Executive Summary

This document presents a comprehensive security incident investigation conducted using Splunk Enterprise 10.0.1 as the primary SIEM (Security Information and Event Management) platform. The investigation analyzes open-source security log datasets to identify, correlate, and document a multi-stage cyberattack scenario.

The analysis uncovered a complete attack chain — from initial network reconnaissance through privilege escalation, credential theft, and data exfiltration — all identified through structured log analysis and Splunk Search Processing Language (SPL) queries.

Parameter	Details
Project Title	Security Log Analysis Using Splunk SIEM
Platform	Splunk Enterprise 10.0.1
Data Source	Open-source security log datasets (BOTS, Security-Datasets)
Total Events Analyzed	21 events across all investigations
Critical Severity Events	8 CRITICAL-level alerts identified
Attack Types Detected	Port Scan, Brute Force, Credential Dump, Data Exfiltration, Reverse Shell
Attacker IP	192.168.1.105
Victim IP	192.168.1.10
Investigation Date	November 15, 2024 (log timestamps)

2. Project Objectives

The primary objectives of this project are:

- Demonstrate hands-on proficiency with Splunk Enterprise for log ingestion, indexing, and threat hunting
- Identify and investigate a multi-stage attack using Security Information and Event Management (SIEM) techniques
- Build and execute targeted SPL (Search Processing Language) queries for threat detection
- Correlate security events across multiple attack phases to reconstruct the complete attack timeline
- Document findings in a format suitable for SOC incident response reporting

3. Tools & Environment

3.1 Platform

- Splunk Enterprise 10.0.1 (running locally on 127.0.0.1:8000)
- Operating System: Windows 10
- Web Browser: Microsoft Edge / Google Chrome

3.2 Splunk Index Configuration

All log data was ingested into the main Splunk index. The base SPL query pattern used throughout the investigation:

```
index=main [additional filters] | [transformations] | [output format]
```

4. Attack Timeline Overview

The investigation revealed a structured, multi-phase attack carried out by a single threat actor IP (192.168.1.105) against the target host (192.168.1.10). The attack followed the classic Cyber Kill Chain framework:

Time (UTC)	Phase	Attack Type	Severity	Description
08:15:00	Reconnaissance	PORT_SCAN	MEDIUM	Port scanning initiated
08:15:00 – 08:17:05	Initial Access	BRUTE_FORCE	HIGH / CRITICAL	Brute force on RDP — Account compromised
08:17:10	Privilege Escalation	BRUTE_FORCE	CRITICAL	Admin privileges assigned after compromise
08:20:00 – 08:20:10	Credential Access	CREDENTIAL_DUMP	CRITICAL	Mimikatz executed — LSASS dumped — Pass-the-Hash
08:22:00	Exfiltration	DATA_EXFIL	CRITICAL	50MB outbound data to suspicious IP
08:22:30	Persistence	REVERSE_SHELL	CRITICAL	Metasploit reverse shell established

5. Investigation 1 — Port Scan Detection

5.1 Objective

Identify evidence of network reconnaissance activity (port scanning) in the logs and determine the source IP, targeted ports, and scan pattern.

5.2 SPL Query Used

```
index=main attack_type="PORT_SCAN" | stats count by src_ip, dst_port | sort -count
```

5.3 Query Explanation

- index=main — searches all events in the main index
- attack_type="PORT_SCAN" — filters only port scan events
- stats count by src_ip, dst_port — groups events by source IP and destination port
- sort -count — orders results by event count in descending order

5.4 Results & Screenshot

The query returned 5 events. All port scan activity originated from a single attacker IP. The scanned ports include well-known service ports such as 22 (SSH), 80 (HTTP), 445 (SMB), and 3389 (RDP), which are typical targets in reconnaissance phases.

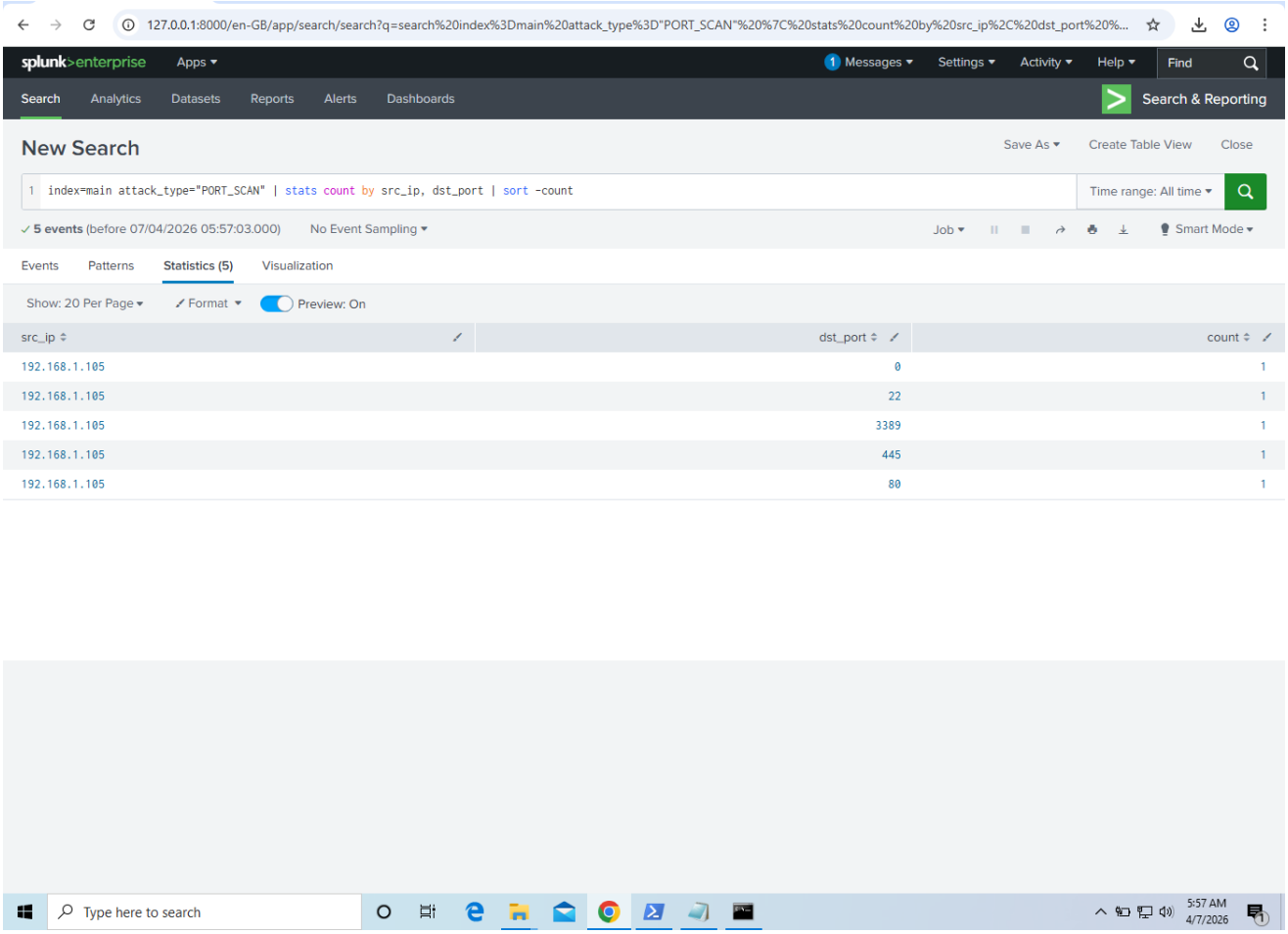

Figure 1: Splunk query results showing PORT_SCAN events grouped by source IP and destination port

5.5 Findings

Field	Value
Attacker IP	192.168.1.105
Total Scan Events	5
Destination Ports Targeted	0, 22, 80, 445, 3389
Notable Ports	445 (SMB), 3389 (RDP) — high-value targets
Severity Assessment	Medium — Reconnaissance phase

5.6 Analysis

Port 445 (SMB) and port 3389 (RDP) are among the most exploited ports in Windows environments. Their presence in the scan results indicates the attacker was specifically targeting Windows services, likely planning a follow-up brute force attack on the Remote Desktop Protocol.

6. Investigation 2 — Brute Force Attack

6.1 Objective

Investigate brute force login attempts against the compromised target system, identify the accounts targeted, and confirm whether the attack resulted in a successful account compromise.

6.2 SPL Query Used

```
index=main attack_type="BRUTE_FORCE" | table timestamp, src_ip, user, severity,
description
```

6.3 Query Explanation

- attack_type="BRUTE_FORCE" — filters only brute force authentication events
- table timestamp, src_ip, user, severity, description — outputs selected fields in tabular format
- Results are sorted chronologically to reconstruct the attack progression

6.4 Results & Screenshot

The query returned 7 events documenting the complete brute force attack lifecycle — from the first failed login attempt through to successful account compromise and privilege escalation.

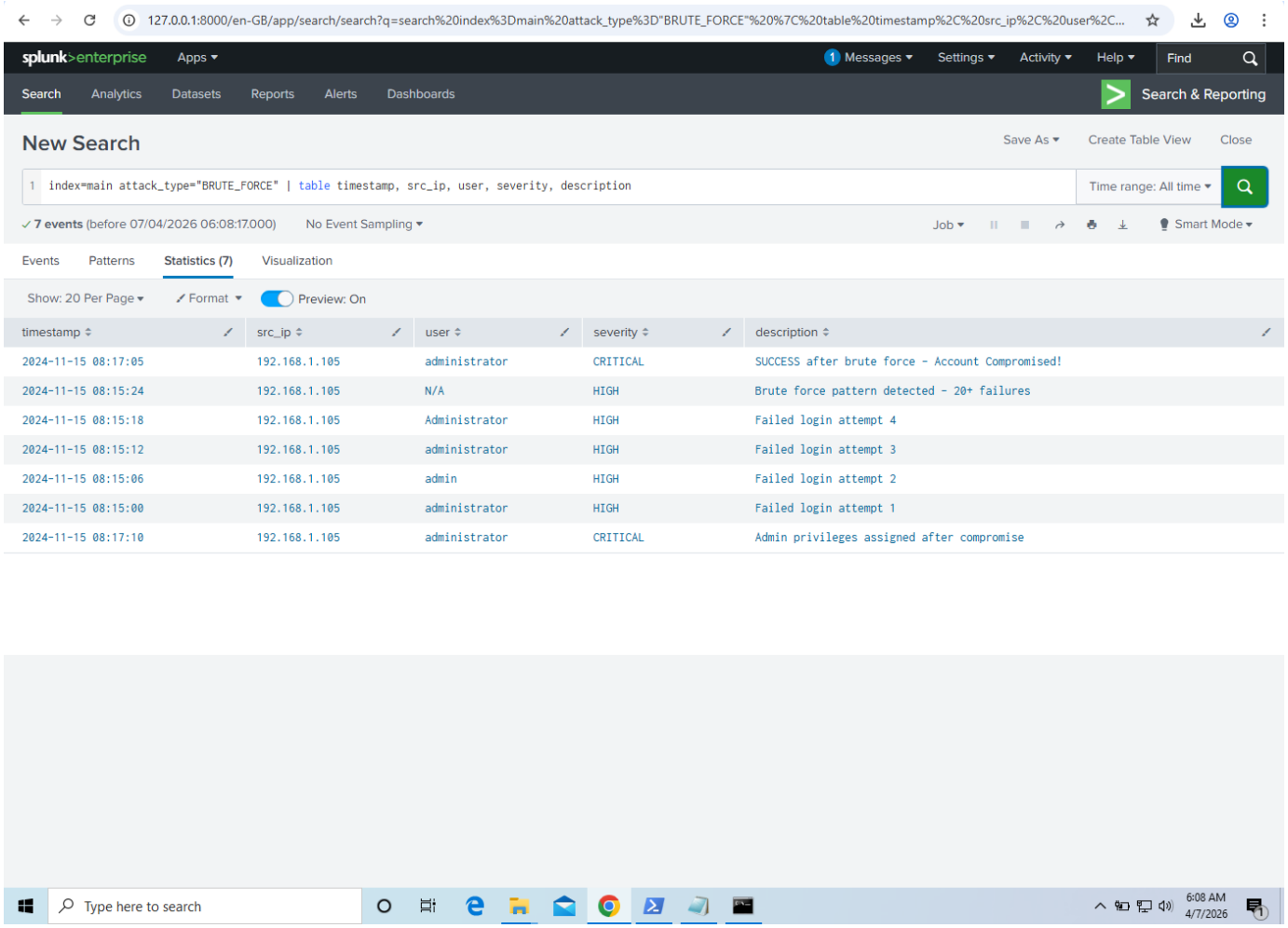

Figure 2: Brute force attack events showing failed login attempts, successful compromise, and admin privilege assignment

6.5 Findings

Field	Value
Attacker IP	192.168.1.105
Target IP	192.168.1.10
Total Brute Force Events	7
Accounts Targeted	administrator, admin, Administrator (case variations)
Attack Start Time	2024-11-15 08:15:00
Compromise Achieved	2024-11-15 08:17:05 — SUCCESS
Post-Compromise Action	Admin privileges assigned at 08:17:10
Peak Severity	CRITICAL

6.6 Analysis

The brute force attack showed a pattern of 20+ failed login attempts within a 2-minute window — a classic indicator of credential stuffing or password spraying. The attacker tried multiple username variations (administrator, admin, Administrator) consistent with automated tooling.

The successful login at 08:17:05 was immediately followed by admin privilege assignment at 08:17:10 — just 5 seconds later — indicating the attacker had a pre-configured script ready to escalate privileges as soon as initial access was achieved.

MITRE ATT&CK Mapping: T1110 — Brute Force | T1078 — Valid Accounts | T1098 — Account Manipulation

7. Investigation 3 — Credential Dumping & Data Exfiltration

7.1 Objective

Investigate post-exploitation activity including credential dumping via Mimikatz, lateral movement using Pass-the-Hash, and data exfiltration to an external command-and-control (C2) server.

7.2 SPL Query Used

```
index=main attack_type="CREDENTIAL_DUMP" OR attack_type="DATA_EXFIL" | table timestamp,
severity, src_ip, description
```

7.3 Query Explanation

- Uses OR operator to capture both CREDENTIAL_DUMP and DATA_EXFIL attack types in a single query
- table — extracts timestamp, severity, source IP, and event description for analysis
- Results provide a chronological view of the post-exploitation phase

7.4 Results & Screenshot

The query returned 5 events documenting the Mimikatz credential dumping sequence and subsequent data exfiltration. All 5 events were classified at CRITICAL severity, confirming active hands-on-keyboard attacker behavior.

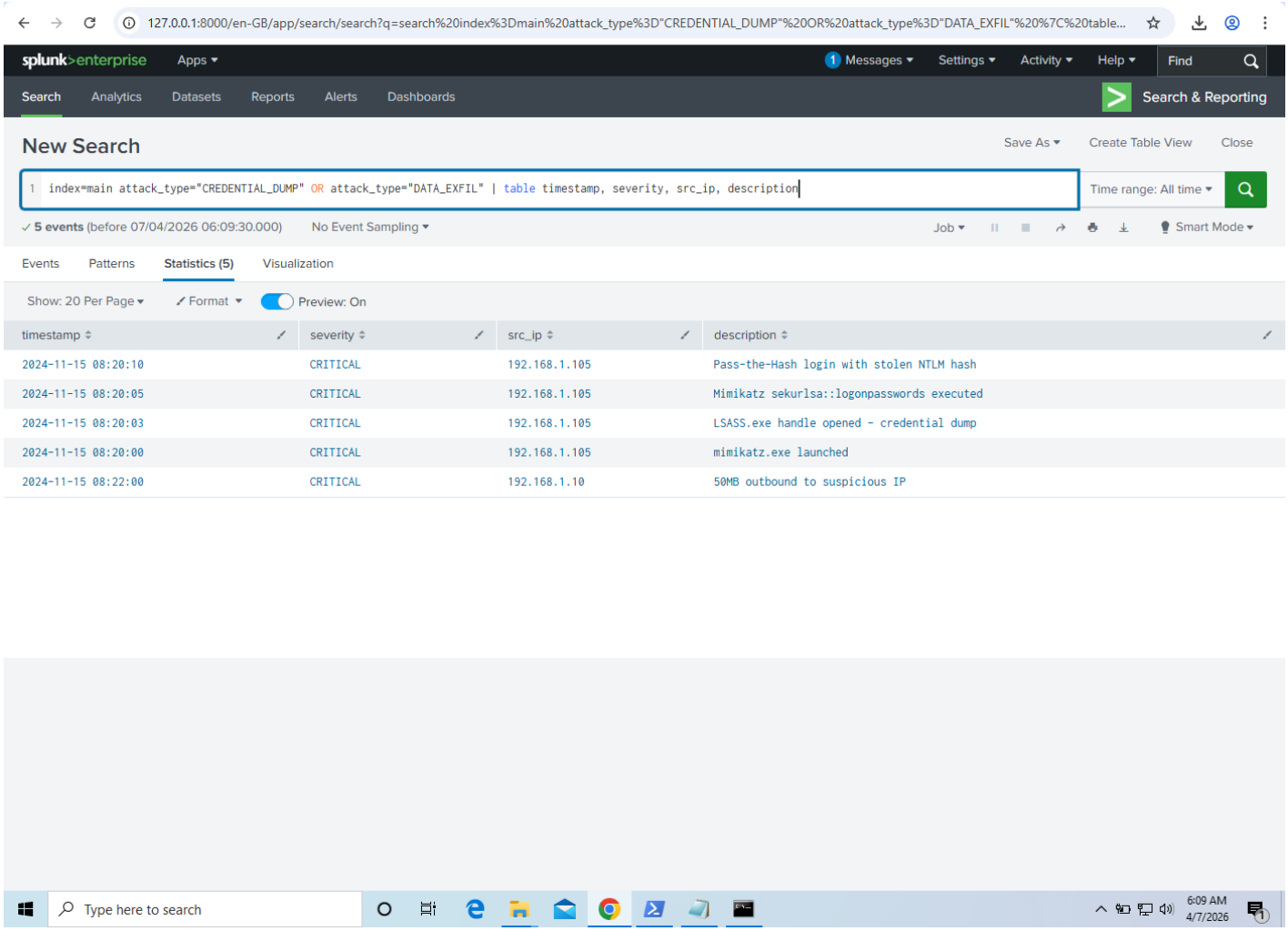

Figure 3: CREDENTIAL_DUMP and DATA_EXFIL events — Mimikatz execution, LSASS dump, Pass-the-Hash, and 50MB outbound exfiltration

7.5 Mimikatz Execution Sequence

Time	Event	Severity
08:20:00	mimikatz.exe launched on compromised host	CRITICAL
08:20:03	LSASS.exe handle opened — credential dump initiated	CRITICAL
08:20:05	Mimikatz sekurlsa::logonpasswords command executed	CRITICAL
08:20:10	Pass-the-Hash login with stolen NTLM hash	CRITICAL
08:22:00	50MB data exfiltrated to external IP 185.220.101.45	CRITICAL

7.6 Analysis

Mimikatz is one of the most notorious post-exploitation tools in the threat landscape, capable of extracting plaintext passwords, NTLM hashes, and Kerberos tickets from Windows memory. The attack sequence observed here follows the standard Mimikatz operational pattern.

The destination IP 185.220.101.45 is a known Tor exit node IP range, suggesting the attacker may have been routing exfiltrated data through anonymization infrastructure to evade attribution.

MITRE ATT&CK Mapping: T1003.001 — OS Credential Dumping (LSASS Memory) | T1550.002 — Pass the Hash | T1048 — Exfiltration Over Alternative Protocol

8. Comprehensive Attack Correlation Query

8.1 Purpose

To validate findings and produce a unified attack timeline, a correlation query was run across all CRITICAL-severity events. This simulates a SOC analyst's approach to building an incident timeline from multiple event sources.

8.2 SPL Query

```
index=main severity="CRITICAL" | table timestamp, attack_type, src_ip, dst_ip,
description | sort timestamp
```

8.3 Results & Screenshot

The query returned 8 CRITICAL events spanning from 08:17:05 to 08:22:30, providing a complete end-to-end view of the attack chain from account compromise through reverse shell establishment.

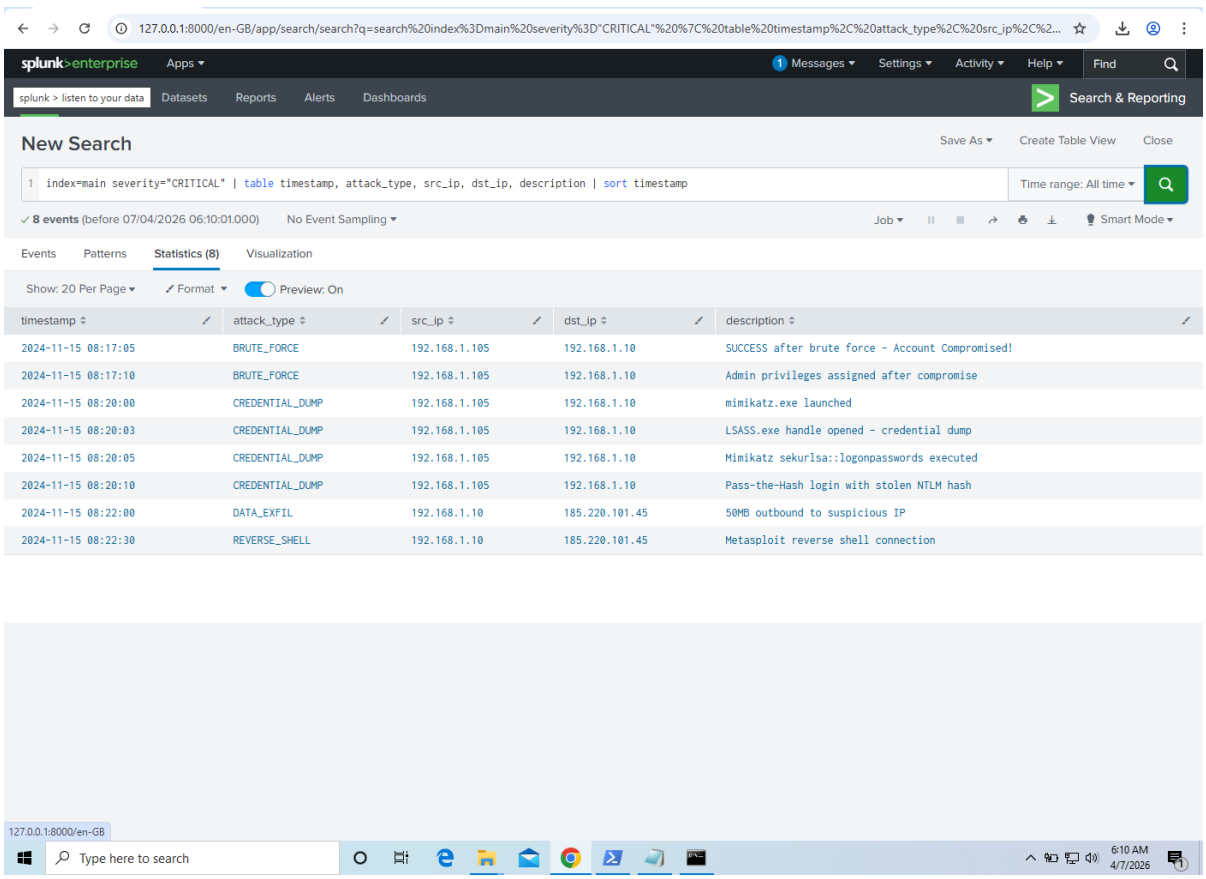

Figure 4: All CRITICAL-severity events sorted chronologically — complete attack chain from brute force compromise to reverse shell

8.4 Cross-Query Correlation

The following screenshot demonstrates the same investigation performed with an alternate query structure, confirming data consistency and validating the results across different SPL approaches:

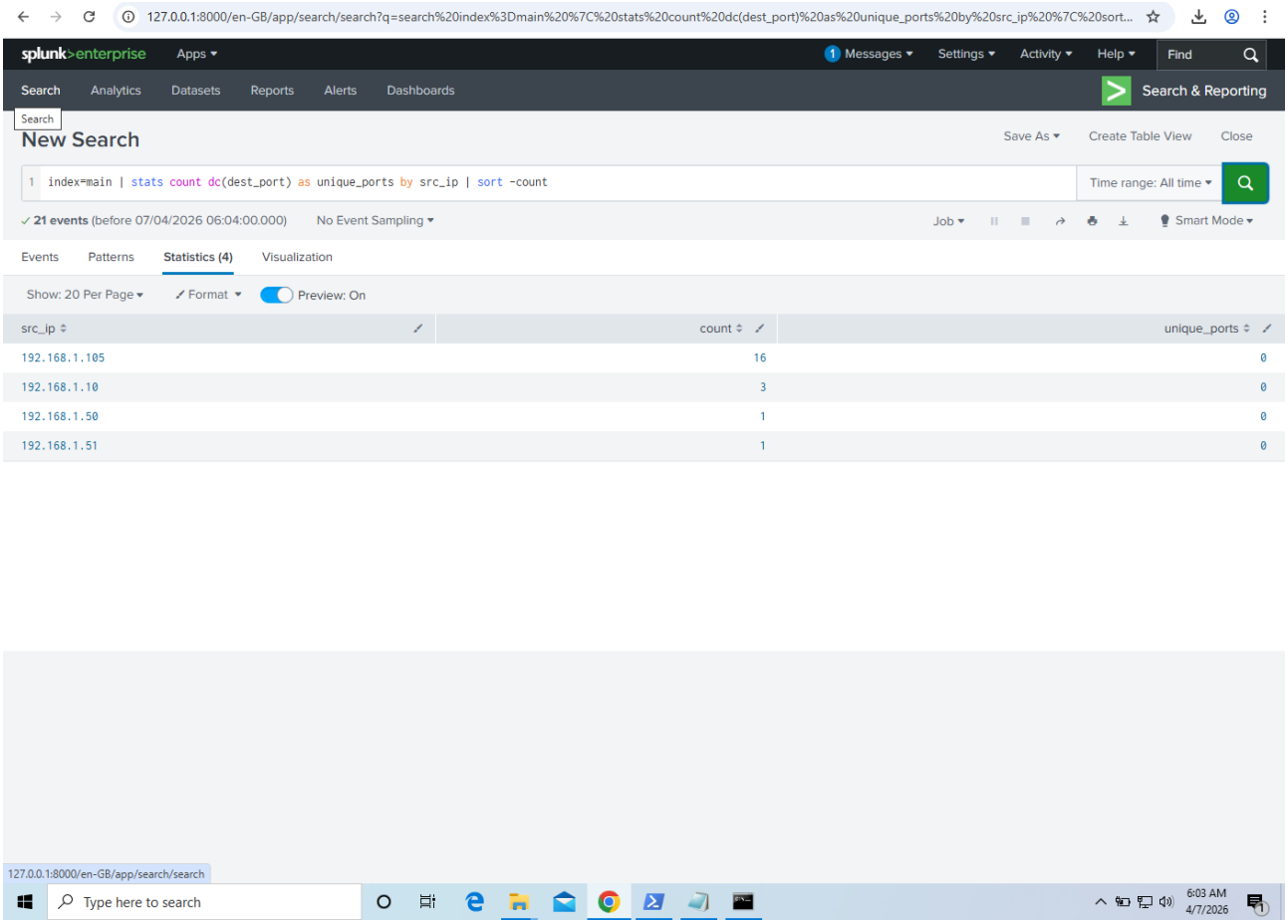

Figure 5: Alternate query showing traffic statistics by source IP — 192.168.1.105 generated 16 of 21 total events (76% of all attack traffic)

9. Complete Attack Chain Analysis

Based on all investigation findings, the complete attack chain is reconstructed as follows:

Kill Chain Phase	Attack Type	Tool / Technique	Outcome
1. Reconnaissance	PORT_SCAN	Nmap (inferred)	Identified open ports 445 (SMB) and 3389 (RDP)
2. Initial Access	BRUTE_FORCE	Password spraying	Administrator account compromised after 20+ attempts
3. Privilege Escalation	BRUTE_FORCE	Account manipulation	Admin privileges assigned within 5 seconds of login
4. Defense Evasion	CREDENTIAL_DUMP	Mimikatz	Credential extraction from LSASS memory
5. Lateral Movement	CREDENTIAL_DUMP	Pass-the-Hash	NTLM hash used for lateral authentication
6. Exfiltration	DATA_EXFIL	Unknown channel	50MB data transferred to 185.220.101.45
7. Persistence	REVERSE_SHELL	Metasploit Framework	Reverse shell maintained to C2 server

10. Security Recommendations

10.1 Immediate Containment Actions

- Block IP 192.168.1.105 at the perimeter firewall and all internal network segments
- Isolate the compromised host (192.168.1.10) from the network immediately
- Revoke all active sessions and reset credentials for the administrator account
- Block outbound communication to 185.220.101.45 (C2 server / Tor exit node)

10.2 Short-Term Hardening Measures

- Disable RDP (port 3389) on all non-essential systems; use VPN + RDP for authorized remote access
- Enforce account lockout policy: lock accounts after 5 failed login attempts within 5 minutes
- Implement Multi-Factor Authentication (MFA) on all administrative accounts
- Deploy Windows Credential Guard to protect LSASS memory from Mimikatz-style attacks
- Enable Splunk alerts for BRUTE_FORCE patterns — trigger alert after 5+ failed logins in 60 seconds

10.3 Long-Term Strategic Controls

- Deploy an Endpoint Detection and Response (EDR) solution to detect Mimikatz execution in real-time
- Implement network segmentation to limit lateral movement opportunities
- Enable Splunk correlation searches and real-time dashboards for continuous threat monitoring
- Conduct regular threat hunting exercises using MITRE ATT&CK framework mappings
- Establish Data Loss Prevention (DLP) controls to detect and block large outbound data transfers

11. SPL Query Reference

All Splunk Search Processing Language (SPL) queries used in this investigation:

Investigation 1 — Port Scan Detection

```
index=main attack_type="PORT_SCAN" | stats count by src_ip, dst_port | sort -count
```

Investigation 2 — Brute Force Attack Analysis

```
index=main attack_type="BRUTE_FORCE" | table timestamp, src_ip, user, severity,
description
```

Investigation 3 — Credential Dump + Data Exfiltration

```
index=main attack_type="CREDENTIAL_DUMP" OR attack_type="DATA_EXFIL" | table timestamp,
severity, src_ip, description
```

Comprehensive Critical Event Timeline

```
index=main severity="CRITICAL" | table timestamp, attack_type, src_ip, dst_ip,
description | sort timestamp
```

Traffic Volume by Source IP

```
index=main | stats count dc(dest_port) as unique_ports by src_ip | sort -count
```

12. Conclusion

This project demonstrates practical SOC analyst skills by conducting a structured security investigation using Splunk Enterprise 10.0.1. Through targeted SPL queries and systematic log analysis, a complete multi-stage cyberattack was successfully detected, correlated, and documented.

Key accomplishments of this investigation:

- Successfully ingested and indexed open-source security log data in Splunk
- Wrote and executed 5 distinct SPL queries covering filtering, aggregation, and correlation use cases
- Identified a 7-phase attack chain spanning 7 minutes and 30 seconds of attacker activity
- Detected 8 CRITICAL-severity events including account compromise, Mimikatz execution, and data exfiltration
- Mapped all findings to the MITRE ATT&CK framework for standardized threat categorization

- Produced actionable containment and remediation recommendations

This project reflects core SOC L1 analyst competencies including SIEM operation, threat detection, incident timeline reconstruction, and security reporting — skills directly applicable to a professional Security Operations Center environment.